

Лабораторная работа №6

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Цель работы

Приобретение практических навыков взаимодействия пользователя с системой посредством командной строки.

Задание

- Определить полное имя домашнего каталога
- Выполнить действия с каталогом /tmp
- Создать и удалить каталоги
- Изучить опции команды ls через man
- Просмотреть описание команд cd, pwd, mkdir, rmdir, rm
- Использовать history для модификации команд

Выполнение лабораторной работы

Определяю полное имя домашнего каталога
командой pwd.

```
avramenkodavid@fedora:~$ pwd  
/home/avramenkodavid
```

Перехожу в каталог /tmp и проверяю
местоположение

```
avramenkodavid@fedora:~$ cd /tmp  
avramenkodavid@fedora:/tmp$ ls
```

Вывожу содержимое каталога /tmp командой ls.

```
avramenkodavid@fedora:/tmp$ ls
007b4a01-ed5a-4cc8-9644-48661e769a40.zip
node-compile-cache
systemd-private-4aaec30b6e1947989b8623b711b7e685-abrtd.service-kkTV9d
systemd-private-4aaec30b6e1947989b8623b711b7e685-chronyd.service-dqqp0g
systemd-private-4aaec30b6e1947989b8623b711b7e685-colord.service-EKcuXL
systemd-private-4aaec30b6e1947989b8623b711b7e685-dbus-broker.service-ps6bgu
systemd-private-4aaec30b6e1947989b8623b711b7e685-fwupd.service-HT00Jw
systemd-private-4aaec30b6e1947989b8623b711b7e685-irqbalance.service-oZ9EtU
systemd-private-4aaec30b6e1947989b8623b711b7e685-ModemManager.service-I3xo3s
systemd-private-4aaec30b6e1947989b8623b711b7e685-passim.service-Th0I92
systemd-private-4aaec30b6e1947989b8623b711b7e685-pcsd.service-xqXCiR
systemd-private-4aaec30b6e1947989b8623b711b7e685-polkit.service-8TkHWq
systemd-private-4aaec30b6e1947989b8623b711b7e685-rtkit-daemon.service-Vl54BG
systemd-private-4aaec30b6e1947989b8623b711b7e685-switcheroo-control.service-WEWANX
systemd-private-4aaec30b6e1947989b8623b711b7e685-systemd-logind.service-JCDMNC
systemd-private-4aaec30b6e1947989b8623b711b7e685-upower.service-UI9lom
```

Вывожу содержимое включая скрытые файлы.

```
avramenkodavid@fedora:/tmp$ ls -l
итого 3872
-rw-r--r--. 1 avramenkodavid avramenkodavid 3963782 авг 23 18:32 007b4a01-ed5a-4cc8-9644-48661e769a40.zip
drwxr-xr-x. 3 avramenkodavid avramenkodavid      60 авг 23 18:38 node-compile-cache
drwx-----. 3 root          root              60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-abrtd.service-kkTV9d
drwx-----. 3 root          root              60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-chronyd.service-dqqp0g
drwx-----. 3 root          root              60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-colord.service-EKcuXL
drwx-----. 3 root          root              60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-dbus-broker.service-ps6bgu
drwx-----. 3 root          root              60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-fwupd.service-HT00Jw
drwx-----. 3 root          root              60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-irqbalance.service-oZ9EtU
drwx-----. 3 root          root              60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-ModemManager.service-I3xo3s
drwx-----. 3 root          root              60 авг 24 19:55 systemd-private-4aaec30b6e1947989b8623b711b7e685-passim.service-Th0I92
drwx-----. 3 root          root              60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-pcsd.service-xqXCiR
drwx-----. 3 root          root              60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-polkit.service-8TkHWq
drwx-----. 3 root          root              60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-rtkit-daemon.service-Vl54BG
drwx-----. 3 root          root              60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-switcheroo-control.service-WEWANX
drwx-----. 3 root          root              60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-systemd-logind.service-JCDMNC
drwx-----. 3 root          root              60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-upower.service-UI9lom
```

Вывожу подробную информацию о файлах.

```
avramenkodavid@fedora:/tmp$ ls -la
.
..
007b4a01-ed5a-4cc8-9644-48661e769a40.zip
.font-unix
.ICE-unix
.iprt-localipc-DRMIPCServer
node-compile-cache
systemd-private-4aaec30b6e1947989b8623b711b7e685-abrtd.service-kkTV9d
systemd-private-4aaec30b6e1947989b8623b711b7e685-chrond.service-dqqp0g
systemd-private-4aaec30b6e1947989b8623b711b7e685-colord.service-EKcuXL
systemd-private-4aaec30b6e1947989b8623b711b7e685-dbus-broker.service-ps6bgu
systemd-private-4aaec30b6e1947989b8623b711b7e685-fwupd.service-HT00Jw
systemd-private-4aaec30b6e1947989b8623b711b7e685-irqbalance.service-oZ9EtU
systemd-private-4aaec30b6e1947989b8623b711b7e685-ModemManager.service-I3xo3s
systemd-private-4aaec30b6e1947989b8623b711b7e685-passim.service-Th0I92
systemd-private-4aaec30b6e1947989b8623b711b7e685-pcscd.service-xqXCiR
systemd-private-4aaec30b6e1947989b8623b711b7e685-polkit.service-8TkHWq
systemd-private-4aaec30b6e1947989b8623b711b7e685-rtkit-daemon.service-Vl54BG
systemd-private-4aaec30b6e1947989b8623b711b7e685-switcheroo-control.service-WEWANX
systemd-private-4aaec30b6e1947989b8623b711b7e685-systemd-logind.service-JCDMNC
systemd-private-4aaec30b6e1947989b8623b711b7e685-upower.service-UI9lom
.wget-hpkp_lock_1000
.wget-hsts_lock_1000
.X0-lock
.X11-unix
.X1-lock
.XIM-unix
```

Комбинированный вывод с указанием типов файлов.

```
avramenkodavid@fedora:/tmp$ ls -la
итого 3880
drwxrwxrwt. 21 root          root          540 авг 24 21:36 .
dr-xr-xr-x.  1 root          root          154 авг 22 17:07 ..
-rw-r--r--.  1 avramenkodavid avramenkodavid 3963782 авг 23 18:32 007b4a81-ed5a-4cc8-9644-48661e769a40.zip
drwxrwxrwt.  2 root          root           40 авг 23 15:05 .font-unix
drwxrwxrwt.  2 root          root           40 авг 23 15:05 .ICE-unix
srw-rw-rw-.  1 root          root            0 авг 23 15:05 .iprt-localipc-DRMlpcServer
drwxr-xr-x.  3 avramenkodavid avramenkodavid   60 авг 23 18:38 node-compile-cache
drwx-----.  3 root          root           60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-abrt.d.service-kkTV9d
drwx-----.  3 root          root           60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-chronyd.service-dqqp0g
drwx-----.  3 root          root           60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-colord.service-EKcuXL
drwx-----.  3 root          root           60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-dbus-broker.service-ps6bgu
drwx-----.  3 root          root           60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-fwupd.service-HT00Jw
drwx-----.  3 root          root           60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-irqbalance.service-oZ9EtU
drwx-----.  3 root          root           60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-ModemManager.service-I3xo3s
drwx-----.  3 root          root           60 авг 24 19:55 systemd-private-4aaec30b6e1947989b8623b711b7e685-passim.service-Th0I92
drwx-----.  3 root          root           60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-pcsd.service-xqXCiR
drwx-----.  3 root          root           60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-polkit.service-8TkHWq
drwx-----.  3 root          root           60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-rtkit-daemon.service-V154BG
drwx-----.  3 root          root           60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-switcheroo-control.service-WEWANX
drwx-----.  3 root          root           60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-systemd-logind.service-JCDMNC
drwx-----.  3 root          root           60 авг 23 15:05 systemd-private-4aaec30b6e1947989b8623b711b7e685-upower.service-UI9lom
-rw-r--r--.  1 avramenkodavid avramenkodavid    0 авг 23 21:46 .wget-hpkp_lck_1000
-rw-r--r--.  1 avramenkodavid avramenkodavid    0 авг 23 21:46 .wget-hsts_lck_1000
-r--r--r--.  1 avramenkodavid avramenkodavid   11 авг 23 15:05 .X0-lock
drwxrwxrwt.  2 root          root           80 авг 23 15:05 .X11-unix
-r--r--r--.  1 avramenkodavid avramenkodavid   11 авг 23 15:05 .X1-lock
drwxrwxrwt.  2 root          root           40 авг 23 15:05 .XIM-unix
```

Проверяю наличие подкаталога cron в /var/spool.

```
avramenkodavid@fedora:/tmp$ ls /var/spool  
abrt  abrt-upload  anacron  at  cron  cups  lpd  mail  plymouth  
avramenkodavid@fedora:/tmp$ cd ~
```

Перехожу в домашний каталог и определяю владельца файлов.

```
avramenkodavid@fedora:~$ ls -l
итого 300
drwxr-xr-x. 1 avramenkodavid avramenkodavid    14 авг 23 21:46 bin
-rw-r--r--. 1 avramenkodavid avramenkodavid   207 авг 23 15:10 git-config.txt
drwxr-xr-x. 1 avramenkodavid avramenkodavid    74 авг 23 19:03 git-extended
-rw-r--r--. 1 avramenkodavid avramenkodavid  8742 авг 23 15:10 gpg-keys-backup.tar.gz
drwxr-xr-x. 1 avramenkodavid avramenkodavid    28 авг 23 15:11 home
-rw-r--r--. 1 avramenkodavid avramenkodavid 282980 авг 23 15:10 installed-packages.txt
-rw-r--r--. 1 avramenkodavid avramenkodavid   3528 авг 23 15:09 ssh-keys-backup.tar.gz
drwxr-xr-x. 1 avramenkodavid avramenkodavid    430 авг 23 15:17 study_2025-2026_os-intro
drwxr-xr-x. 1 avramenkodavid avramenkodavid     0 авг 23 15:00 Видео
drwxr-xr-x. 1 avramenkodavid avramenkodavid     0 авг 23 15:00 Документы
drwxr-xr-x. 1 avramenkodavid avramenkodavid     0 авг 23 15:00 Загрузки
drwxr-xr-x. 1 avramenkodavid avramenkodavid    50 авг 23 19:29 Изображения
drwxr-xr-x. 1 avramenkodavid avramenkodavid     0 авг 23 15:00 Музыка
drwxr-xr-x. 1 avramenkodavid avramenkodavid     0 авг 23 15:00 Общедоступные
drwxr-xr-x. 1 avramenkodavid avramenkodavid     0 авг 23 15:00 'Рабочий стол'
drwxr-xr-x. 1 avramenkodavid avramenkodavid     0 авг 23 15:00 Шаблоны
```

Создаю каталог newdir и подкаталог morefun.

```
avramenkodavid@fedora:~$ mkdir newdir
avramenkodavid@fedora:~$ ls
bin          gpg-keys-backup.tar.gz  newdir      Видео      Изображения  'Рабочий стол'
git-config.txt  home                   ssh-keys-backup.tar.gz  Документы  Музыка      Шаблоны
git-extended  installed-packages.txt  study_2025-2026_os-intro  Загрузки  Общедоступные
avramenkodavid@fedora:~$ mkdir newdir/morefun
avramenkodavid@fedora:~$ ls newdir/
morefun
```

Создаю три каталога одной командой и удаляю их.

```
avramenkodavid@fedora:~$ mkdir letters memos misk
avramenkodavid@fedora:~$ ls
bin                gpg-keys-backup.tar.gz  letters  newdir            Видео      Изображения  'Рабочий стол'
git-config.txt     home                   memos    ssh-keys-backup.tar.gz  Документы  Музыка       Шаблоны
git-extended       installed-packages.txt  misk     study_2025-2026_os-intro  Загрузки   Общедоступные
avramenkodavid@fedora:~$ rmdir letters memos misk
avramenkodavid@fedora:~$ ls
bin                gpg-keys-backup.tar.gz  newdir            Видео      Изображения  'Рабочий стол'
git-config.txt     home                   ssh-keys-backup.tar.gz  Документы  Музыка       Шаблоны
git-extended       installed-packages.txt  study_2025-2026_os-intro  Загрузки   Общедоступные
avramenkodavid@fedora:~$ rm newdir
rm: невозможно удалить 'newdir': Это каталог
avramenkodavid@fedora:~$ ls
bin                gpg-keys-backup.tar.gz  newdir            Видео      Изображения  'Рабочий стол'
git-config.txt     home                   ssh-keys-backup.tar.gz  Документы  Музыка       Шаблоны
git-extended       installed-packages.txt  study_2025-2026_os-intro  Загрузки   Общедоступные
```

Команда `rm` без опций не может удалить каталог.

```
avramenkodavid@fedora:~$ rm newdir  
rm: невозможно удалить 'newdir': Это каталог
```

Удаляю morefun и newdir с опцией -r.

```
avramenkodavid@fedora:~$ rm -r newdir/morefun  
avramenkodavid@fedora:~$ ls newdir/
```

Определяю опцию для рекурсивного просмотра.

```
avramenkodavid@fedora:~ — man ls

-l      use a long listing format

-L, --dereference
        when showing file information for a symbolic link, show information for the file the link references rather than for the link itself

-m      fill width with a comma separated list of entries

-n, --numeric-uid-gid
        like -l, but list numeric user and group IDs

-N, --literal
        print entry names without quoting

-o      like -l, but do not list group information

-p, --indicator-style=slash
        append / indicator to directories

-g, --hide-control-chars
        print ? instead of nongraphic characters

--show-control-chars
        show nongraphic characters as-is; the default, unless program is 'ls' and output is a terminal

-Q, --quote-name
        enclose entry names in double quotes

--quoting-style=WORD
        use quoting style WORD for entry names: literal, locale, shell, shell-always, shell-escape, shell-escape-always, c, escape (overrides QUOTING_STYLE environment variable)

-r, --reverse
        reverse order while sorting

-R, --recursive
        list subdirectories recursively
```

Демонстрирую рекурсивный просмотр /etc/cron.d.

```
avramenkodavid@fedora:~$ man ls  
avramenkodavid@fedora:~$ man ls
```

Определяю опции для сортировки по времени, man ls — опция -l

```
avramenkodavid@fedora:~ — man ls

-l      use a long listing format
-L, --dereference
        when showing file information for a symbolic link, show information for the file the link references rather than for the link itself

-m      fill width with a comma separated list of entries

-n, --numeric-uid-gid
        like -l, but list numeric user and group IDs

-N, --literal
        print entry names without quoting

-o      like -l, but do not list group information

-p, --indicator-style=slash
        append / indicator to directories

-q, --hide-control-chars
        print ? instead of nongraphic characters

--show-control-chars
        show nongraphic characters as-is; the default, unless program is 'ls' and output is a terminal

-Q, --quote-name
        enclose entry names in double quotes

--quoting-style=WORD
        use quoting style WORD for entry names: literal, locale, shell, shell-always, shell-escape, shell-escape-always, c, escape (overrides QUOTING_STYLE environment variable)

-r, --reverse
        reverse order while sorting

-R, --recursive
        list subdirectories recursively

Manual page ls(1) line 97 (press h for help or q to quit)
```

Определяю опции для сортировки по времени, man ls — опция -t

```
avramenkodavid@fedora:~ — man ls

use quoting style WORD for entry names: literal, locale, shell, shell-always, shell-escape, shell-escape-always, c, escape (overrides QUOT-
ING_STYLE environment variable)

-r, --reverse
    reverse order while sorting

-R, --recursive
    list subdirectories recursively

-s, --size
    print the allocated size of each file, in blocks

-S
    sort by file size, largest first

--sort=WORD
    change default 'name' sort to WORD: none (-U), size (-S), time (-t), version (-v), extension (-X), name, width

--time=WORD
    select which timestamp used to display or sort; access time (-u): atime, access, use; metadata change time (-c): ctime, status; modified time
    (default): mtime, modification; birth time: birth, creation; with -l, WORD determines which time to show; with --sort=time, sort by WORD
    (newest first)

--time-style=TIME_STYLE
    time/date format with -l; see TIME_STYLE below

-t
    sort by time, newest first; see --time
    |
-T, --tabsize=COLS
    assume tab stops at each COLS instead of 8

-u
    with -lt: sort by, and show, access time; with -l: show access time and sort by name; otherwise: sort by access time, newest first

-U
    do not sort directory entries

-v
    natural sort of (version) numbers within text

-w, --width=COLS
    Manual page ls(1) line 125 (press h for help or q to quit)
```

Вывожу содержимое домашнего каталога с
сортировкой по времени.

```
avramenkodavid@fedora:~$ ls -lt
```

итого 300

drwxr-xr-x.	1	avramenkodavid	avramenkodavid	0	авг 24 21:46	newdir
drwxr-xr-x.	1	avramenkodavid	avramenkodavid	14	авг 23 21:46	bin
drwxr-xr-x.	1	avramenkodavid	avramenkodavid	50	авг 23 19:29	Изображения
drwxr-xr-x.	1	avramenkodavid	avramenkodavid	74	авг 23 19:03	git-extended
drwxr-xr-x.	1	avramenkodavid	avramenkodavid	430	авг 23 15:17	study_2025-2026_os-intro
drwxr-xr-x.	1	avramenkodavid	avramenkodavid	28	авг 23 15:11	home
-rw-r--r--.	1	avramenkodavid	avramenkodavid	282980	авг 23 15:10	installed-packages.txt
-rw-r--r--.	1	avramenkodavid	avramenkodavid	207	авг 23 15:10	git-config.txt
-rw-r--r--.	1	avramenkodavid	avramenkodavid	8742	авг 23 15:10	gpg-keys-backup.tar.gz
-rw-r--r--.	1	avramenkodavid	avramenkodavid	3528	авг 23 15:09	ssh-keys-backup.tar.gz
drwxr-xr-x.	1	avramenkodavid	avramenkodavid	0	авг 23 15:00	Видео
drwxr-xr-x.	1	avramenkodavid	avramenkodavid	0	авг 23 15:00	Документы
drwxr-xr-x.	1	avramenkodavid	avramenkodavid	0	авг 23 15:00	Загрузки
drwxr-xr-x.	1	avramenkodavid	avramenkodavid	0	авг 23 15:00	Музыка
drwxr-xr-x.	1	avramenkodavid	avramenkodavid	0	авг 23 15:00	Общедоступные
drwxr-xr-x.	1	avramenkodavid	avramenkodavid	0	авг 23 15:00	'Рабочий стол'
drwxr-xr-x.	1	avramenkodavid	avramenkodavid	0	авг 23 15:00	Шаблоны

```
avramenkodavid@fedora:~$
```

Просматриваю справку по команде cd.

```
avramenkodavid@fedora:~ — man cd

BASH_BUILTINS(1)                                General Commands Manual                                BASH_BUILTINS(1)

NAME
:, ., [, alias, bg, bind, break, builtin, caller, cd, command, compgen, complete, compopt, continue, declare, dirs, disown, echo, enable, eval, exec,
exit, export, false, fc, fg, getopts, hash, help, history, jobs, kill, let, local, logout, mapfile, popd, printf, pushd, pwd, read, readarray, read-
only, return, set, shift, shopt, source, suspend, test, times, trap, true, type, typeset, ulimit, umask, unalias, unset, wait - bash built-in com-
mands, see bash(1)

BASH BUILTIN COMMANDS
Unless otherwise noted, each builtin command documented in this section as accepting options preceded by - accepts -- to signify the end of the op-
tions. The :, true, false, and test/[ builtins do not accept options and do not treat -- specially. The exit, logout, return, break, continue, let,
and shift builtins accept and process arguments beginning with - without requiring --. Other builtins that accept arguments but are not specified as
accepting options interpret arguments beginning with - as invalid options and require -- to prevent this interpretation.

: [arguments]
    No effect; the command does nothing beyond expanding arguments and performing any specified redirections. The return status is zero.

. [-p path] filename [arguments]
source [-p path] filename [arguments]
    The . command (source) reads and execute commands from filename in the current shell environment and returns the exit status of the last com-
    mand executed from filename.

    If filename does not contain a slash, . searches for it. If the -p option is supplied, . treats path as a colon-separated list of directories
    in which to find filename; otherwise, . uses the entries in PATH to find the directory containing filename. filename does not need to be exe-
    cutable. When bash is not in posix mode, it searches the current directory if filename is not found in PATH, but does not search the current
    directory if -p is supplied. If the sourcepath option to the shopt builtin command is turned off, . does not search PATH.

    If any arguments are supplied, they become the positional parameters when filename is executed. Otherwise the positional parameters are un-
    changed.

    If the -T option is enabled, . inherits any trap on DEBUG; if it is not, any DEBUG trap string is saved and restored around the call to ., and
    . unsets the DEBUG trap while it executes. If -T is not set, and the sourced file changes the DEBUG trap, the new value persists after . com-
    pletes. The return status is the status of the last command executed from filename (0 if no commands are executed), and non-zero if filename
    is not found or cannot be read.

alias [-p] [name=value] ...]
    With no arguments or with the -p option, alias prints the list of aliases in the form alias name=value on standard output. When arguments are
    supplied, define an alias for each name whose value is given. A trailing space in value causes the next word to be checked for alias substitu-
    tion when the alias is expanded during command parsing. For each name in the argument list for which no value is supplied, print the name and
    value of the alias name. alias returns true unless a name is given (without a corresponding =value) for which no alias has been defined.

bg [jobspec ...]
    Resume each suspended job jobspec in the background, as if it had been started with &. If jobspec is not present, the shell uses its notion of
    the current job. bg jobspec returns 0 unless run when job control is disabled or, when run with job control enabled, any specified jobspec was
    not found or was started without job control.

bind [-m keymap] [-lsvSX]
bind [-m keymap] [-q function] [-u function] [-r keyseq]
bind [-m keymap] -f filename

Manual page cd(1) line 1 (press h for help or q to quit)
```

Просматриваю справку по команде pwd.

```
avramenkodavid@fedora:~ — man pwd

PWD(1)                                User Commands

NAME
    pwd - print name of current/working directory

SYNOPSIS
    pwd [OPTION]...

DESCRIPTION
    Print the full filename of the current working directory.

    -L, --logical
        use PWD from environment, even if it contains symlinks

    -P, --physical
        resolve all symlinks

    --help
        display this help and exit

    --version
        output version information and exit

    If no option is specified, -P is assumed.

    Your shell may have its own version of pwd, which usually supersedes the version described here. Please refer to your
    documentation for details about the options it supports.

AUTHOR
    Written by Jim Meyering.

REPORTING BUGS
    Report bugs to: bug-coreutils@gnu.org
    GNU coreutils home page: <https://www.gnu.org/software/coreutils/>
    General help using GNU software: <https://www.gnu.org/gethelp/>

Manual page pwd(1) line 1 (press h for help or q to quit)
```

Просматриваю справку по команде mkdir.

```
avramenkodavid@fedora:~ — man mkdir

MKDIR(1) User Commands

NAME
  mkdir - make directories

SYNOPSIS
  mkdir [OPTION]... DIRECTORY...

DESCRIPTION
  Create the DIRECTORY(ies), if they do not already exist.

  Mandatory arguments to long options are mandatory for short options too.

  -m, --mode=MODE
      set file mode (as in chmod), not a=rwx - umask

  -p, --parents
      no error if existing, make parent directories as needed, with their file modes unaffected by any

  -v, --verbose
      print a message for each created directory

  -Z
      set SELinux security context of each created directory to the default type

  --context[=CTX]
      like -Z, or if CTX is specified then set the SELinux or SMACK security context to CTX

  --help
      display this help and exit

  --version
      output version information and exit

AUTHOR
  Written by David MacKenzie.

Manual page mkdir(1) line 1 (press h for help or q to quit)
```

Просматриваю справку по команде rmdir.

```
avramenkodavid@fedora:~ — man rmdir

RMDIR(1)                                User Commands

NAME
    rmdir - remove empty directories

SYNOPSIS
    rmdir [OPTION]... DIRECTORY...

DESCRIPTION
    Remove the DIRECTORY(ies), if they are empty.

    --ignore-fail-on-non-empty
        ignore each failure to remove a non-empty directory

    -p, --parents
        remove DIRECTORY and its ancestors; e.g., 'rmdir -p a/b' is similar to 'rmdir a/b a'

    -v, --verbose
        output a diagnostic for every directory processed

    --help
        display this help and exit

    --version
        output version information and exit

AUTHOR
    Written by David MacKenzie.

REPORTING BUGS
    Report bugs to: bug-coreutils@gnu.org
    GNU coreutils home page: <https://www.gnu.org/software/coreutils/>
    General help using GNU software: <https://www.gnu.org/gethelp/>
    Report any translation bugs to <https://translationproject.org/team/>

Manual page rmdir(1) line 1 (press h for help or q to quit)
```

Просматриваю справку по команде rm.

```
avramenkodavid@fedora:~ — man rm
RM(1)                                User Commands
NAME
  rm - remove files or directories
SYNOPSIS
  rm [OPTION]... [FILE]...
DESCRIPTION
  This manual page documents the GNU version of rm.  rm removes each specified file.  By default, it does
  not remove directories.

  If the -i or --interactive=once option is given, and there are more than three files or the -r, -R, or -d
  options are given, rm prompts the user for whether to proceed with the entire operation.  If the response is not affirmative, the operation is aborted.

  Otherwise, if a file is unwritable, standard input is a terminal, and the -f or --force option is not given, or the --interactive=always option is given, rm prompts the user for whether to remove the file.  If the response is not affirmative, the file is skipped.
OPTIONS
  Remove (unlink) the FILE(s).

  -f, --force
  .....,
  ignore nonexistent files and arguments, never prompt

  -i
  .....,
  prompt before every removal

  -I
  .....,
  prompt once before removing more than three files, or when removing recursively; less intrusive than -i;
  useful for protection against most mistakes

  --interactive[=WHEN]
  .....,
  prompt according to WHEN: never, once (-I), or always (-i); without WHEN, prompt always

  --one-file-system
  .....,
  Manual page rm(1) line 1 (press h for help or q to quit)
```

Вывожу историю команд.

```
avramenkodavid@fedora:~$ history
1  lsblk
2  cd /run/media/avramenkodavid/MIREX
3  ls -la
4  cp ./ssh-keys-backup.tar.gz ~/
5  cp ./gpg-keys-backup.tar.gz ~/
6  cp ./git-config.txt ~/
7  cp ./installed-packages.txt ~/
8  cd ~
9  tar -xzf ssh-keys-backup.tar.gz
10 chmod 700 ~/.ssh
11 tar -tzf ssh-keys-backup.tar.gz | head -10
12 ls -la ~/.ssh/
13 sudo tar -xzf ssh-keys-backup.tar.gz -C /
14 ls -la ~/.ssh/
15 ls -la ~/.gnupg/
16 sudo tar -xzf gpg-keys-backup.tar.gz -C /
17 ls -la ~/.gnupg/
18 gpg --list-secret-keys
19 sudo dnf install git gh -y
20 git config --global user.name "Avramenko David Andreevich"
21 git config --global user.email "avramenkodavid22@gmail.com"
22 git config --global user.signingkey 277D80FE26A062E1
23 git config --global commit.gpgsign true
24 git config --global core.quotepath false
25 git config --global init.defaultBranch master
26 git config --global --list
27 ssh -T git@github.com
28 git clone git@github.com:avramenkodavid22-dotcom/study_2025-2026_os-intro.git
29 cd study_2025-2026_os-intro
30 git log --show-signature -1
31 git status
32 cd ~
33 cd ~/study_2025-2026_os-intro
34 mkdir -p labs/lab02
35 cd labs/lab02
36 touch report.md
37 gedit report.md
38 -cd
39 touch report.md
40 mkdir -p labs/lab02
41 cd labs/lab02
42 touch report.md
43 cd ~
44 cd ~/study_2025-2026_os-intro/labs/lab02
```

Модификация команды из истории

```
avramenkodavid@fedora:~$ !3:s/ls/ls -la/
ls -la -la
итого 344
drwx-----. 1 avramenkodavid avramenkodavid 1124 авг 24 21:45 .
drwxr-xr-x. 1 root root 28 авг 23 15:00 ..
-rw-----. 1 avramenkodavid avramenkodavid 4767 авг 24 21:35 .bash_history
-rw-r--r--. 1 avramenkodavid avramenkodavid 18 янв 16 2026 .bash_logout
-rw-r--r--. 1 avramenkodavid avramenkodavid 144 янв 16 2026 .bash_profile
-rw-r--r--. 1 avramenkodavid avramenkodavid 684 авг 23 18:38 .bashrc
drwxr-xr-x. 1 avramenkodavid avramenkodavid 14 авг 23 21:46 bin
drwx-----. 1 avramenkodavid avramenkodavid 482 авг 23 21:27 .cache
drwxr-xr-x. 1 avramenkodavid avramenkodavid 408 авг 23 22:32 .config
-rw-r--r--. 1 avramenkodavid avramenkodavid 191 авг 23 15:16 .gitconfig
-rw-r--r--. 1 avramenkodavid avramenkodavid 207 авг 23 15:10 git-config.txt
drwxr-xr-x. 1 avramenkodavid avramenkodavid 74 авг 23 19:03 git-extended
drwx-----. 1 avramenkodavid avramenkodavid 158 авг 23 21:37 .gnupg
-rw-r--r--. 1 avramenkodavid avramenkodavid 8742 авг 23 15:10 gpg-keys-backup.tar.gz
drwxr-xr-x. 1 avramenkodavid avramenkodavid 28 авг 23 15:11 home
-rw-r--r--. 1 avramenkodavid avramenkodavid 282980 авг 23 15:10 installed-packages.txt
drwxr-xr-x. 1 avramenkodavid avramenkodavid 20 авг 23 15:00 .local
drwxr-xr-x. 1 avramenkodavid avramenkodavid 0 авг 24 21:46 newdir
drwx-----. 1 avramenkodavid avramenkodavid 62 авг 23 21:39 .password-store
drwx-----. 1 avramenkodavid avramenkodavid 84 авг 22 22:06 .ssh
-rw-r--r--. 1 avramenkodavid avramenkodavid 3528 авг 23 15:09 ssh-keys-backup.tar.gz
drwxr-xr-x. 1 avramenkodavid avramenkodavid 430 авг 23 15:17 study_2025-2026_os-intro
-rw-r-----. 1 avramenkodavid avramenkodavid 5 авг 23 15:05 .vboxclient-clipboard-tty2-control.pid
-rw-r-----. 1 avramenkodavid avramenkodavid 5 авг 23 15:05 .vboxclient-clipboard-tty2-service.pid
-rw-r-----. 1 avramenkodavid avramenkodavid 5 авг 23 15:05 .vboxclient-draganddrop-tty2-control.pid
-rw-r-----. 1 avramenkodavid avramenkodavid 5 авг 23 15:05 .vboxclient-hostversion-tty2-control.pid
-rw-r-----. 1 avramenkodavid avramenkodavid 5 авг 23 15:05 .vboxclient-seamless-tty2-control.pid
drwxr-xr-x. 1 avramenkodavid avramenkodavid 0 авг 23 15:00 Видео
drwxr-xr-x. 1 avramenkodavid avramenkodavid 0 авг 23 15:00 Документы
drwxr-xr-x. 1 avramenkodavid avramenkodavid 0 авг 23 15:00 Загрузки
drwxr-xr-x. 1 avramenkodavid avramenkodavid 50 авг 23 19:29 Изображения
drwxr-xr-x. 1 avramenkodavid avramenkodavid 0 авг 23 15:00 Музыка
drwxr-xr-x. 1 avramenkodavid avramenkodavid 0 авг 23 15:00 Общедоступные
drwxr-xr-x. 1 avramenkodavid avramenkodavid 0 авг 23 15:00 'Рабочий стол'
drwxr-xr-x. 1 avramenkodavid avramenkodavid 0 авг 23 15:00 Шаблоны
```

Выводы

В ходе выполнения лабораторной работы приобрёл практические навыки взаимодействия с системой посредством командной строки. Освоил команды навигации (`cd`, `pwd`), просмотра каталогов (`ls`), создания и удаления каталогов (`mkdir`, `rm`), получения справки (`man`) и работы с историей команд (`history`).